



TRIOS CYBER

LABORATORY PRACTICE 20

Windows Wazuh Agent Deployment and Enrollment Verification

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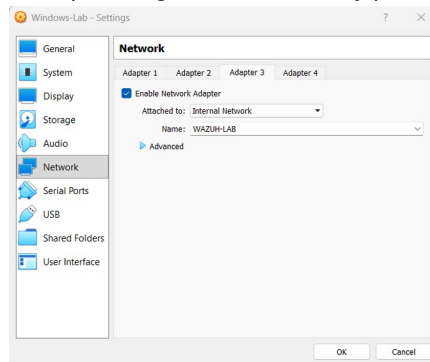
1. Laboratory Objective and Environment

The Day 20 activity deployed a Wazuh agent on the Windows laboratory virtual machine and enrolled it with the existing Wazuh server. The validation focused on the private WAZUH-LAB path, Windows endpoint addressing, connectivity to the Wazuh agent ports, local service startup, and final confirmation that the endpoint was Active in Wazuh.

Element	Observed value	Purpose / result
Wazuh server	10.10.20.10/24	Manager and enrollment services
Windows endpoint	10.10.20.30/24	Windows system monitored by Wazuh
Internal network	WAZUH-LAB	Private VirtualBox communication path

2. VirtualBox Internal Network Configuration

The Windows-Lab virtual machine was connected to the same VirtualBox Internal Network used by the Wazuh server. The additional adapter was enabled and attached to WAZUH-LAB, providing a direct laboratory path between the endpoint and the server.



3. Wazuh Server Address and Service Readiness

On the Wazuh server, interface enp0s9 was assigned 10.10.20.10/24. The Wazuh indexer, manager, Filebeat, and dashboard were restarted, and each service returned active before the Windows endpoint was enrolled.

```
matthew@trioscyber:~$ sudo ip link set enp0s9 up
sudo ip addr flush dev enp0s9
sudo ip addr add 10.10.20.10/24 dev enp0s9
ip -br a | grep enp0s9
[sudo] password for matthew:
enp0s9                UP                10.10.20.10/24

matthew@trioscyber:~$ sudo systemctl restart wazuh-indexer
sleep 15
sudo systemctl restart wazuh-manager
sudo systemctl restart filebeat
sudo systemctl restart wazuh-dashboard
sleep 15
sudo systemctl is-active wazuh-indexer
sudo systemctl is-active wazuh-manager
sudo systemctl is-active filebeat
sudo systemctl is-active wazuh-dashboard
active
active
active
active
```

4. Windows Endpoint Network Configuration

PowerShell was used to identify the Windows network adapters and configure the WAZUH-LAB interface. Ethernet 3 received the static address 10.10.20.30/24. The final ipconfig output confirmed the address while the other adapters retained their existing settings.

```
PS C:\Windows\system32> Get-NetAdapter

Name      InterfaceDescription      ifIndex Status      MacAddress      LinkSpeed
-----
Ethernet  Intel(R) PRO/1000 MT Desktop Adapter      6 Up          08-00-27-70-9C-56      1 Gbps
Ethernet 2 Intel(R) PRO/1000 MT Desktop Adapter #2    10 Up         08-00-27-DF-DD-72      1 Gbps
Ethernet 3 Intel(R) PRO/1000 MT Desktop Adapter #3    11 Up         08-00-27-5D-5E-39      1 Gbps

PS C:\Windows\system32> New-NetIPAddress -InterfaceAlias "Ethernet 3" -IPAddress 10.10.20.30 -PrefixLength 24

IPAddress      : 10.10.20.30
InterfaceIndex  : 11
InterfaceAlias  : Ethernet 3
AddressFamily   : IPv4
Type            : Unicast
PrefixLength    : 24
PrefixOrigin    : Manual
SuffixOrigin    : Manual
AddressState    : Tentative
ValidLifetime   : Infinite ([TimeSpan]):MaxValue
PreferredLifetime : Infinite ([TimeSpan]):MaxValue
SkipSource      : False
PolicyStore     : ActiveStore

PS C:\Windows\system32> ipconfig

Configuración IP de Windows

Adaptador de Ethernet Ethernet:

    Sufixo DNS específico para la conexión. . . :
    Vínculo dirección IPv6 local. . . : fe80::b1d1:85f1:401d:deec%10
    Dirección IPv4. . . : 10.10.10.20
    Máscara de subred . . . : 255.255.255.0
    Puerta de enlace predeterminada . . . :

Adaptador de Ethernet Ethernet 2:

    Sufixo DNS específico para la conexión. . . :
    Vínculo dirección IPv6 local. . . : fe80::3de9:3d45:8a8f:cf4b%11
    Dirección IPv4. . . : 10.0.3.15
    Máscara de subred . . . : 255.255.255.0
    Puerta de enlace predeterminada . . . : 10.0.3.2

Adaptador de Ethernet Ethernet 3:

    Sufixo DNS específico para la conexión. . . :
    Vínculo dirección IPv6 local. . . : fe80::3de9:3d45:8a8f:cf4b%11
    Dirección IPv4. . . : 10.10.20.30
    Máscara de subred . . . : 255.255.255.0
    Puerta de enlace predeterminada . . . :

PS C:\Windows\system32>
```

5. Connectivity Verification to the Wazuh Server

The Windows endpoint tested the Wazuh server directly with Test-NetConnection. TCP ports 1514 and 1515 both returned TcpTestSucceeded: True from source address 10.10.20.30, confirming reachability to the communication and enrollment services at 10.10.20.10.

```

PS C:\Windows\system32> Test-NetConnection 10.10.20.10 -Port 1514
ComputerName      : 10.10.20.10
RemoteAddress     : 10.10.20.10
RemotePort        : 1514
InterfaceAlias    : Ethernet 3
SourceAddress     : 10.10.20.30
TcpTestSucceeded  : True

PS C:\Windows\system32> Test-NetConnection 10.10.20.10 -Port 1515
ComputerName      : 10.10.20.10
RemoteAddress     : 10.10.20.10
RemotePort        : 1515
InterfaceAlias    : Ethernet 3
SourceAddress     : 10.10.20.30
TcpTestSucceeded  : True

```

Validation	Observed result	Meaning
TCP 1514	TcpTestSucceeded: True	Agent communication reachable
TCP 1515	TcpTestSucceeded: True	Enrollment service reachable

6. Dashboard Access through the Existing SSH Tunnel

The existing SSH local-port-forwarding method was reused to access the Wazuh Dashboard from the host. Local port 8443 was forwarded through the SSH session on host port 2222 to 127.0.0.1:443 on the Wazuh server.

```

PS C:\Users\User> ssh -N -L 8443:127.0.0.1:443 -p 2222 matthew@127.0.0.1
matthew@127.0.0.1's password:

```

7. Windows Agent Deployment from the Wazuh Dashboard

The Deploy new agent workflow was opened from Endpoints. The server address was set to 10.10.20.10 and the agent name to Windows-Agent. Wazuh generated the Windows installation command used on the endpoint.

For additional systems and architectures, please check our [documentation](#).

Server address:
This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address

10.10.20.10

☒ Remember server address

Optional settings:
By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name:

Windows-Agent

The agent name must be unique. It can't be changed once the agent has been enrolled.

Select one or more existing groups:

Windows-LAB

Run the following commands to download and install the agent:

```
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.7-1.msi -OutFile $env:tmp\wazuh-agent; msixexec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER="10.10.20.10" WAZUH_AGENT_NAME="Windows-Agent"
```

Requirements

8. Agent Installation and Local Service Verification

The Wazuh 4.14.7 Windows MSI package was installed with WAZUH_MANAGER and WAZUH_REGISTRATION_SERVER set to 10.10.20.10 and WAZUH_AGENT_NAME set to Windows-Agent. The Wazuh service was started, and Get-Service reported wazuhsvc with status Running.

```

PS C:\Windows\system32> cd C:\
PS C:\> cd C:\Temp
PS C:\Temp> msixexec.exe /i wazuh-agent-4.14.7-1.msi WAZUH_MANAGER="10.10.20.10" WAZUH_REGISTRATION_SERVER="10.10.20.10" WAZUH_AGENT_NAME="Windows-Agent"
PS C:\Temp> NET START WazuhSvc
El servicio de Wazuh está iniciándose.
El servicio de Wazuh se ha iniciado correctamente.

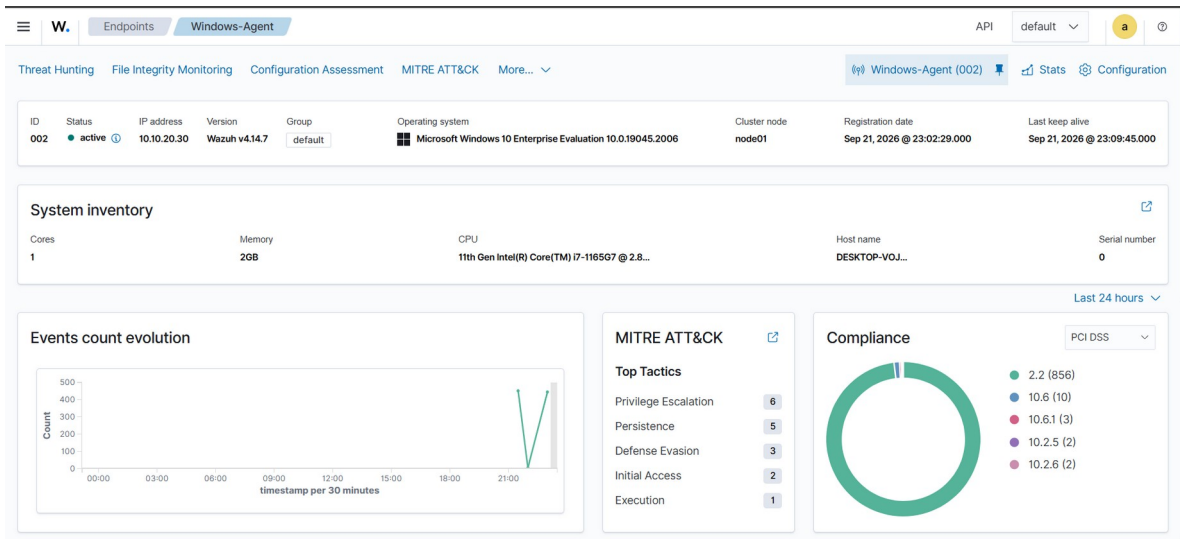
PS C:\Temp> Get-Service wazuhsvc

Status  Name      DisplayName
-----
Running wazuhsvc Wazuh

```

9. Final Windows Agent Registration and Active Status

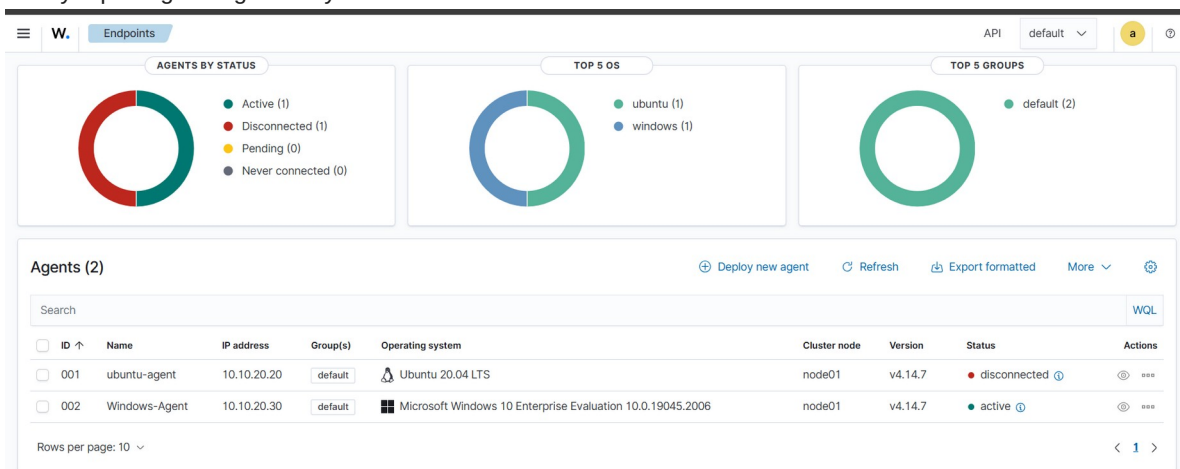
After the local Wazuh service started, the Endpoints view showed agent ID 002 as Windows-Agent at 10.10.20.30. The endpoint reported Wazuh v4.14.7, Microsoft Windows 10 Enterprise Evaluation, cluster node node01, group default, and status Active, confirming successful communication with the manager.



Element	Observed value	Purpose / result
Agent ID / Name	002 / Windows-Agent	Registered Windows endpoint
IP address	10.10.20.30	Expected WAZUH-LAB address
Version / Group	v4.14.7 / default	Installed agent and effective group
Local service	wazuhsvc - Running	Agent service operational
Final status	Active	Successful manager-agent communication

10. Consolidated Endpoint Inventory

The final Endpoints summary showed two registered agents in the default group: Windows-Agent (ID 002, 10.10.20.30) was Active and ubuntu-agent (ID 001, 10.10.20.20) was Disconnected at the time of capture. This confirmed that the Windows endpoint remained actively reporting during the Day 20 validation.



11. Conclusions

The Windows Wazuh agent deployment was completed successfully. Windows-Lab communicated with the Wazuh server through WAZUH-LAB, TCP ports 1514 and 1515 were reachable, and local verification confirmed wazuhsvc was Running. Wazuh showed Windows-Agent at 10.10.20.30 with Active status. The consolidated Endpoints view also preserved the previous ubuntu-agent registration, listed as Disconnected at the time of capture. This satisfies the Day 20 requirement to install, enroll, locally verify, and confirm an active Windows Wazuh endpoint.